

Frequency-Domain Anomaly Detection for Encrypted Traffic in Industrial Control Systems

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Abstract—Industrial control systems (ICSs) are becoming increasingly interconnected, rendering them susceptible to cyber attacks. Timely detection of anomalies in encrypted data flows is crucial for ensuring the reliability and security of ICS. Although deep learning-based anomaly detection methods have made significant strides, their implementation in point-by-point mapping paradigms often necessitates a tradeoff between feature representation and computational efficiency, especially in resource-constrained environments. In addition, these methods face challenges with class imbalance and limited labeled anomaly data. To address these challenges, we propose a frequency-domain anomaly detection (FreAD) framework specifically tailored for encrypted data flows in ICS. FreAD utilizes a frequency-domain feature fusion encoding module to capture global temporal dependencies. An anomaly scoring network integrates a small amount of labeled data along with pseudolabeled data generated from the modified Z-score algorithm, effectively addressing class imbalance. Furthermore, the frequency-domain deviation prior module can alleviate the contraction of interclass distances during unsupervised training. Extensive experiments demonstrate that FreAD significantly surpasses other state-of-the-art anomaly detection algorithms.

Index Terms—Encrypted traffic, frequency-domain deviation, frequency-domain feature fusion, semi-supervised anomaly detection.

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I. INTRODUCTION

INDUSTRIAL control systems (ICSs) are increasingly interconnected through modern communication networks. The increasing complexity of ICS and the proliferation of industrial Internet of Things devices have brought new vulnerabilities to cyber attacks, leading to increasingly serious network security issues [1], especially in the detection of anomalies in encrypted traffic [2]. Since traditional detection schemes cannot directly analyze the abnormal activities contained in the encrypted payload content [3], such as data breaches, unauthorized access, or even sabotage [4], therefore, timely and accurate detection of anomalies in encrypted traffic is crucial to ensure the reliability and security of ICS and mitigate the risk of catastrophic failures or interruptions.

Many studies have explored the challenge of detecting anomalies in encrypted traffic within ICS through various methodologies, encompassing traditional statistical techniques [5], [6], [7], [20], [26] as well as deep learning (DL)-based approaches [8], [9], [10], [11], [12], [13], [14], [15], [16]. Traditional methods typically rely on handcrafted expert features, which often struggle to capture intricate temporal dependencies and are susceptible to challenges posed by high-dimensional data. In contrast, DL-based techniques, such as multilayer perceptrons (MLPs) [10], [11], autoencoders (AEs) [12], [13], [14], [15], and convolutional neural networks (CNNs) [16], [17], have demonstrated significant success in deciphering complex patterns from encrypted flows. Notably, recent advancements in Transformer-based models [18], [19] exhibit enhanced capabilities in modeling global temporal dependencies. However, DL-based methods generally necessitate considerable computational resources for both training and inference, presenting challenges for resource-constrained ICS environments. Furthermore, point-by-point mapping techniques employed in certain architectures may still encounter representation information bottlenecks when processing high-frequency sequential data. These limitations, coupled with inherent challenges, such as class imbalance and the scarcity of labeled anomaly data, underscore the need for alternative solutions that effectively balance performance with operational feasibility.

To address these challenges, we introduce FreAD, a frequency-domain semi-supervised anomaly detection framework for encrypted traffic in ICS. Specifically, FreAD leverages a frequency-domain feature fusion encoding module (F³EM) to extract features from encrypted data flows, overcoming the

limitations of traditional DL-based methods that rely on point-by-point mapping for anomaly detection. It effectively captures global temporal dependencies and mitigates information bottlenecks by harnessing the global nature of frequency-domain transformations. Subsequently, an anomaly scoring network integrates a small amount of anomaly labeled data and pseudolabeled data generated by the modified Z-score (MZS) algorithm to tackle class imbalance and limited labeled anomaly data in ICS. In addition, a frequency-domain deviation prior module (FDPM) is incorporated during unsupervised training to mitigate the narrowing of interclass distances by identifying and emphasizing frequency-domain deviations. Extensive experimental results show that FreAD outperforms other state-of-the-art anomaly detection algorithms in terms of area under the receiver operating characteristic curve (AUCROC), area under the precision-recall curve (AUCPR), and F1-score across three diverse datasets.

The contributions of this work are presented as follows.

- 1) We propose a novel frequency-domain semisupervised anomaly detection framework, named FreAD, which addresses the challenges of encrypted network traffic anomaly detection in ICS by leveraging frequency-domain features and prior knowledge about anomalies.
- 2) We develop an F³EM to extract frequency-domain features from encrypted data flows, effectively alleviating the representation information bottleneck associated with traditional DL-based point-by-point mapping anomaly detection methods.
- 3) We establish an anomaly scoring network that integrates a small amount of labeled anomaly data and pseudolabeled data generated by the MZS algorithm, addressing class imbalance and the scarcity of labeled anomaly data in ICS.
- 4) We design an innovative FDPM in unsupervised training that effectively tackles abnormal data deviations while enhancing the framework's robustness against transient anomalies.

The rest of this article is organized as follows. Section II briefly reviews related works. Section III presents the details of the proposed FreAD framework. Section IV compares FreAD with state-of-the-art algorithms and performs ablation experiments. Finally, Section V concludes this article.

II. RELATED WORKS

The security and reliability of ICS have become critical concerns as these systems become increasingly interconnected through modern communication networks. This increased connectivity has introduced significant network security challenges, particularly in detecting anomalies within encrypted network traffic. Researchers have explored various categories of approaches to tackle this issue.

Traditional statistical methods typically rely on manually crafted features extracted from encrypted network data flows [5], [6], [7], [20]. These approaches encompass density estimation techniques [8], clustering methods [20], and probability estimation [6], [26]. While these techniques are interpretable and computationally efficient, they often struggle to capture complex

temporal dependencies in network flows and are vulnerable to the challenges of high-dimensional data. For example, density estimation methods may inadequately model intricate relationships between time steps, while clustering approaches frequently rely on predefined assumptions about data distribution, which limits their adaptability to dynamic network environments.

In contrast, DL-based anomaly detection methods have shown promising results in uncovering complex patterns and relationships within encrypted traffic flows. These methods encompass MLP models [10], [11], AE approaches [12], [13], [14], [15], and CNN frameworks [16], [17]. These models harness the robust feature extraction capabilities of artificial neural networks to analyze high-dimensional data and uncover underlying patterns. However, they face challenges in modeling global temporal dependencies and may encounter representation bottlenecks in point-by-point mappings. Recent transformer-based models [18], [19] are particularly effective at capturing these dependencies, yet they often involve tradeoffs regarding training time and resource consumption, which are common in DL techniques. For instance, OmniAnomaly [18], a variational AE-based approach, excels in modeling intricate dependencies in multivariate time series but struggles with computational efficiency and scalability, particularly in high-dimensional ICS environments. Similarly, the AnomalyTransformer [19], which utilizes self-attention mechanisms to capture long-range dependencies, necessitates prior knowledge of the dataset's anomaly ratio for effective training, limiting its applicability in resource-constrained settings. These DL-based methods also share common limitations, including sensitivity to hyperparameter tuning, high computational complexity, and difficulties in addressing the variability inherent in ICS data.

In light of these challenges, frequency analysis methods for anomaly detection have emerged as a complementary approach. Specifically, the fast Fourier transform (FFT) has become a popular alternative for analyzing time-series data, particularly in detecting anomalies in encrypted traffic. FFT is effective at capturing global temporal dependencies and identifying periodic patterns, making it valuable for spotting anomalies that may be missed in the time domain [30]. However, FFT has its limitations; a significant drawback is its inability to effectively capture transient anomalies due to the assumption of signal stationarity. To address this issue, researchers have suggested alternative transformations, such as the short-time Fourier transform (STFT) [31] and wavelet transform [32]. STFT provides localized analysis through windowed FFT, but it can suffer from time-frequency tradeoffs, which may limit its ability to accurately capture rapid changes in the signal. Meanwhile, wavelet transform offers multiresolution analysis to capture both transient and periodic anomalies; however, it can be computationally intensive and may require careful selection of wavelet functions to achieve optimal results. Despite these alternatives, FFT remains a favored choice for many applications due to its computational efficiency and simplicity.

To overcome these challenges, the proposed FreAD in this work aims to address the challenges of encrypted network traffic anomaly detection in ICS by leveraging frequency-domain analysis and prior knowledge about anomalies. Specifically, FreAD

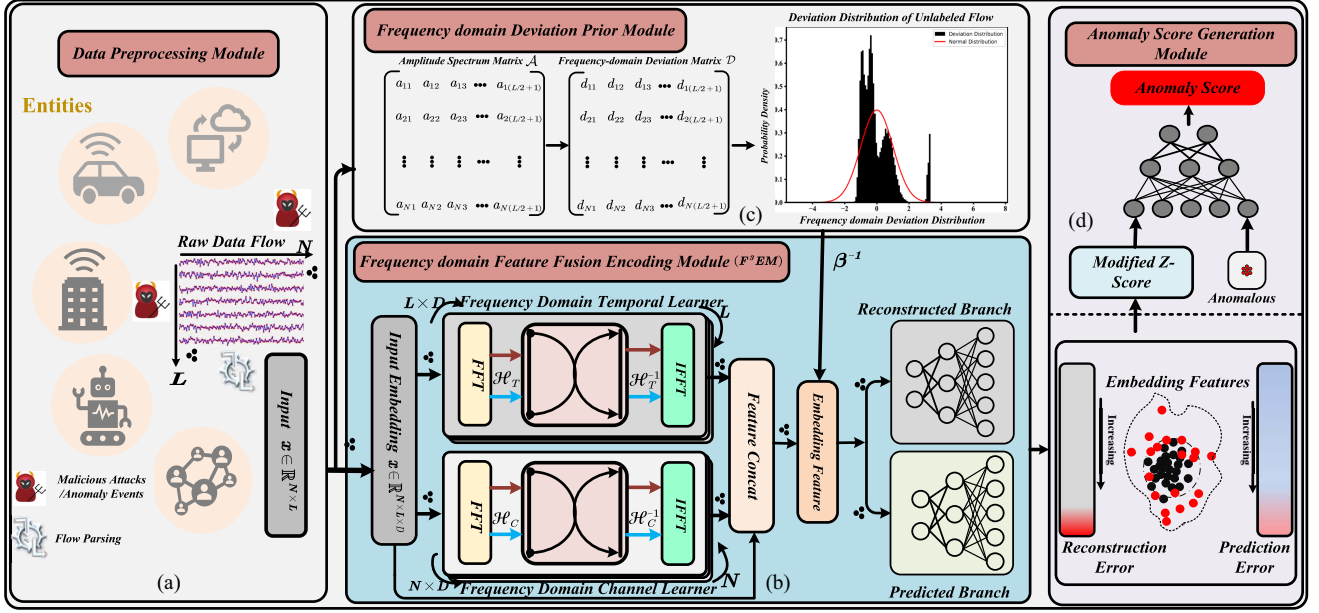


Fig. 1. Details of FreAD: A comprehensive FreAD comprising four parts, i.e., (a) data preprocessing module, (b) F^3EM , (c) FDPM, and (d) ASGM.

incorporates an F^3EM to extract relevant features from encrypted data flows, thereby addressing the representation information bottleneck commonly found in traditional DL-based anomaly detection methods and effectively capturing global temporal dependencies. Subsequently, an anomaly scoring network combines a limited set of labeled anomaly data with pseudolabeled data generated by the MZS algorithm, which helps to overcome class imbalance and the lack of labeled anomaly data in ICS. Furthermore, an FDPM is integrated into the unsupervised training process to enhance the detection of abrupt changes and transient anomalies in encrypted traffic, while mitigating the issue of reduced interclass distances and addressing the limitations of conventional Fourier transforms.

III. FREAD FRAMEWORK

A. Overview of FreAD

An overview of FreAD is shown in Fig. 1. Specifically, to address the shortcomings of traditional DL-based models in capturing global temporal dependencies, FreAD employs an F^3EM to extract comprehensive frequency-domain features. This is followed by the FDPM to mitigate the impact of anomalies during unsupervised training. Finally, the ASGM leverages prior knowledge to learn a scoring function that distinguishes between normal and anomalous data points, enabling accurate anomaly detection and outlier identification.

B. Frequency-Domain Feature Fusion Encoding Module

The F^3EM comprises a frequency-domain feature fusion encoder (F^3E) and an extended decoder. The encoder consists of three components: a frequency-domain temporal learner (FDTL) branch, a frequency domain channel learner (FDCL)

branch, and a residual connection to the original input, the extended decoder has two parts: a reconstruction decoding branch and a prediction decoding branch.

1) *Frequency-Domain Feature Fusion Encoder (F^3E)*: The F^3E processes the original data $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N]^T \in \mathbb{R}^{N \times (L+l)}$, where $N = N_U + N_A$ is the total number of data points, L is the input dimension, and l is the dimension of the predicted output. The $\mathbf{X}$ consists of unlabeled data $\mathbf{X}_U \in \mathbb{R}^{N_U \times (L+l)}$ and a small set of labeled anomalies $\mathbf{X}_A \in \mathbb{R}^{N_A \times (L+l)}$. In addition, $\mathbf{X}^L \in \mathbb{R}^{N \times L}$ represents the input features, while $\mathbf{X}^l \in \mathbb{R}^{N \times l}$ corresponds to the sequence dimensions $\{L+1, L+2, \dots, L+l\}$. The F^3E process can be decomposed into several stages, including the following.

Input embedding: The input $\mathbf{X}^L \in \mathbb{R}^{N \times L}$ is embedded into a higher dimensional space $\mathbf{X}' \in \mathbb{R}^{B \times L \times C}$, where B is the batch size, L is the sequence length, and C is the number of channels ($C = 1$). The embedding operation can be expressed as

$$\mathbf{X}'' = \mathbf{X}' \otimes \mathbf{W}_e \quad (1)$$

where $\mathbf{W}_e \in \mathbb{R}^{1 \times D}$ is the embedding weight, D is the expanded embedding dimension, and $\otimes$ denotes the elementwise multiplication with broadcasting. Specifically, the input $\mathbf{X}^L \in \mathbb{R}^{N \times L}$ is expanded to $\mathbf{X}' \in \mathbb{R}^{B \times L \times C}$ to capture channel information, and then permuted to $\mathbb{R}^{B \times C \times L}$ for subsequent operations. To enhance computational efficiency, the tensor is expanded to $\mathbb{R}^{B \times C \times L \times 1}$ and multiplied by the embedding weight $\mathbf{W}_e \in \mathbb{R}^{1 \times D}$, allowing for shared weights and bias. As a result, the embedded output $\mathbf{X}''$ with dimensions $\mathbb{R}^{B \times C \times L \times D}$, improving the model's ability to capture complex patterns while maintaining efficiency.

Frequency-domain temporal learner: The FDTL transforms the input $\mathbf{X}'' \in \mathbb{R}^{B \times C \times L \times D}$ into the frequency domain, which the process begins with the application of the FFT along the temporal dimension L , projecting the input into the frequency

domain

$$\mathbf{F}_t = \mathbb{H}_t(\mathbf{X}'') \in \mathbb{C}^{B \times C \times (L/2+1) \times D} \quad (2)$$

where $\mathbb{H}_t(\cdot)$ denotes the FFT operation with “ortho” normalization in the temporal dimension, revealing the spectral characteristics of the input across the temporal dimension. After obtaining the frequency domain representation of temporal features, the model employs a critical operation known as the frequency domain MLP (FreMLP), which can be mathematically formulated as the function $\mathcal{F}_{\text{FreMLP}}$

$$\mathbf{F}'_t = \mathcal{F}_{\text{FreMLP}}(\mathbf{F}_t) \in \mathbb{C}^{B \times C \times (L/2+1) \times D} \quad (3)$$

where $\mathbf{F}'_t$ maintains the same dimension as the input, but contains enhanced spectral features learned from the frequency-domain representation. The inverse FFT (IFFT) is applied to return the data to the original domain, completing the spectral transformation

$$\mathbf{E}'_t = \mathbb{H}_t^{-1}(\mathbf{F}'_t) \in \mathbb{R}^{B \times C \times L \times D} \quad (4)$$

where $\mathbb{H}_t^{-1}(\cdot)$ denotes the IFFT operation with “ortho” normalization in the temporal dimension. The computational complexity of the FDTL operation is primarily influenced by the FFT, which operates with a complexity of $O(L \log L)$. This efficiency allows the FDTL to reveal and amplify hidden patterns in the input data, facilitating a deeper understanding of complex temporal dynamics.

FDCL: The FDCL operates on the input $\mathbf{X}'' \in \mathbb{R}^{B \times C \times L \times D}$. First, the input is transposed along the sequence and embedding dimensions to align the channels for spectral analysis, resulting in

$$\mathbf{X}''' = \text{permute}(\mathbf{X}'', \text{dims} = [0, 2, 1, 3]) \in \mathbb{R}^{B \times L \times C \times D} \quad (5)$$

which rearrangement prepares the input for a spectral analysis. The FDCL, then applies the FFT along the channel dimension C , projecting the input into the frequency domain

$$\mathbf{F}_c = \mathbb{H}_c(\mathbf{X}''') \in \mathbb{C}^{B \times L \times (C/2+1) \times D} \quad (6)$$

where $\mathbb{H}_c(\cdot)$ represents the FFT operation with “ortho” normalization in the channel dimension. This process reveals the spectral characteristics of the input across the channel dimension. After obtaining the frequency-domain representation of the channel features, the FDCL employs $\mathcal{F}_{\text{FreMLP}}$, which enhances the spectral features

$$\mathbf{F}'_c = \mathcal{F}_{\text{FreMLP}}(\mathbf{F}_c) \in \mathbb{C}^{B \times L \times (C/2+1) \times D} \quad (7)$$

where $\mathbf{F}'_c$ maintains the same dimension as the input, but contains enhanced spectral features learned from the frequency-domain representation. The IFFT is applied to return the data to the original domain, completing the FDCL operation

$$\mathbf{E}'_c = \mathbb{H}_c^{-1}(\mathbf{F}'_c) \in \mathbb{R}^{B \times L \times C \times D} \quad (8)$$

where $\mathbb{H}_c^{-1}(\cdot)$ denotes the IFFT operation with “ortho” normalization in the channel dimension. The computational complexity of the FDCL operation is primarily influenced by the FFT, which operates with a complexity of $O(N \log N)$. This efficiency allows the FDCL to uncover and amplify hidden spectral patterns

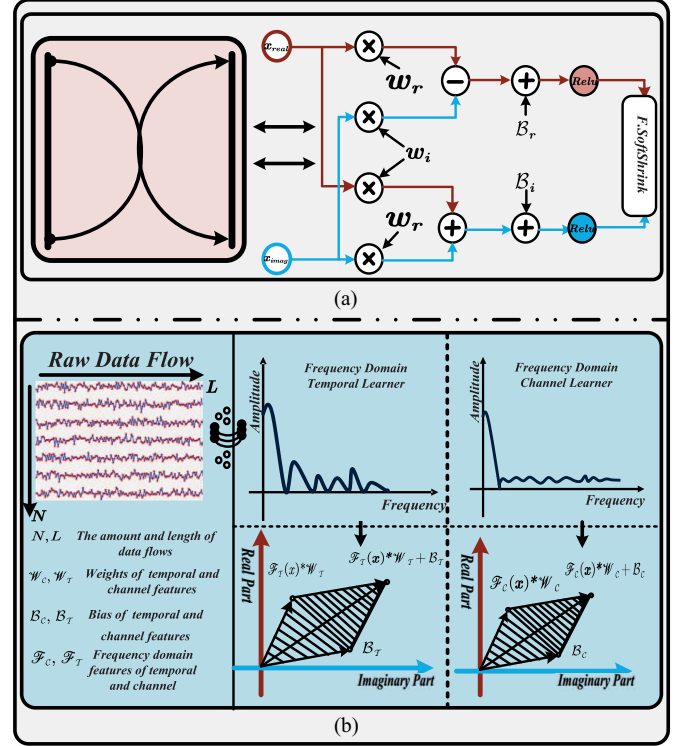


Fig. 2. (a) Schematic diagram illustrating the operational principles of FreMLP. (b) Explanation of the application of FreMLP in FDCL and FDTL.

in the channel dimension, facilitating a deeper understanding of complex channelwise dynamics.

FreMLP: The FreMLP, a key component in both FDTL and FDCL, is shown in Fig. 2, facilitating spectral feature extraction in the frequency domain: Let $\mathbf{X}'' \in \mathbb{C}^{B \times C \times (L/2+1) \times D}$ be the input to the FreMLP, which start by decomposing the input into its real and imaginary components

$$\mathbf{X}''_r = \Re(\mathbf{X}'') \in \mathbb{R}^{B \times C \times (L/2+1) \times D} \quad (9)$$

$$\mathbf{X}''_i = \Im(\mathbf{X}'') \in \mathbb{R}^{B \times C \times (L/2+1) \times D} \quad (10)$$

where $\mathbf{X}''_r$ and $\mathbf{X}''_i$ represent the real part $\Re(\cdot)$ and the imaginary part $\Im(\cdot)$ of $\mathbf{X}''$, respectively. Next, we employ two sets of weight matrices, $\mathbf{W}_R$ and $\mathbf{W}_I$, to process the real and imaginary components, respectively

$$\mathbf{W}_R \in \mathbb{R}^{D \times D}, \mathbf{W}_I \in \mathbb{R}^{D \times D}. \quad (11)$$

The FreMLP performs complex-valued linear transformations, producing outputs from the real and imaginary components as follows:

$$\mathbf{O}_r = \text{Softshrink}(\text{Relu}(\mathbf{X}''_r \mathbf{W}_R - \mathbf{X}''_i \mathbf{W}_I), \lambda) \quad (12)$$

$$\mathbf{O}_i = \text{Softshrink}(\text{Relu}(\mathbf{X}''_i \mathbf{W}_R + \mathbf{X}''_r \mathbf{W}_I), \lambda) \quad (13)$$

where $\mathbf{O}_r$ and $\mathbf{O}_i$ denote the real and imaginary outputs, respectively, and λ is the sparsity threshold applied in the Softshrink function. Finally, the real and imaginary outputs are combined

to form the complex-valued output $\mathbf{F}'$

$$\mathbf{F}' = \mathbf{O}_r + j\mathbf{O}_i \in \mathbb{C}^{B \times C \times (L/2+1) \times D}. \quad (14)$$

The FreMLP layer takes the input $\mathbf{X}''$ and applies a series of complex-valued linear transformations to produce the output $\mathbf{F}'$ that captures enhanced spectral features crucial for the subsequent processing.

Feature fusion: We extract temporal features $\mathbf{E}'_t$ and channel features $\mathbf{E}'_c$ from FDTL and FDCL, respectively. The feature fusion process is expressed as

$$\mathbf{X}^* = \mathbf{E}'_t + \mathbf{E}'_c + \mathbf{X}'' \in \mathbb{R}^{B \times C \times L \times D}. \quad (15)$$

Specifically, the original input $\mathbf{X}''$ is utilized as a residual connection to construct the fusion feature, where it serves as a foundation for integrating the temporal and channel features. Next, a linear transformation is applied to the fused features

$$\mathbf{X}^{**} = \mathcal{L}(\mathbf{X}^*) \in \mathbb{R}^{B \times L \times C} \quad (16)$$

where $\mathcal{L}(\cdot)$ implemented as a fully connected layer. Finally, the output is obtained by applying the Relu function

$$\mathbf{Z} = \text{Relu}(\mathcal{FC}(\mathbf{X}^{**})) \in \mathbb{R}^{B \times z \times C} \quad (17)$$

where $\mathcal{FC}(\cdot)$ is a fully connected layer and z indicates the dimension of the output feature space. The final output $\mathbf{Z}$ emerges from this feature fusion process, seamlessly integrating the features from both the FDTL and FDCL within a cohesive mathematical framework. Consequently, the overall complexity of the F³E is $O(N \log N + L \log L)$, predominantly influenced by the FFT operations employed for input data processing. This complexity underscores the model's adeptness in managing spectral transformations while effectively capturing intricate patterns.

2) Extended Decoder: The extended decoder comprises two key components: the RecDecoder and the PreDecoder. These decoders operate on the encoded representation $\mathbf{Z} \in \mathbb{R}^{B \times z \times C}$ produced by the F³E. The RecDecoder can be succinctly expressed as

$$\tilde{\mathbf{X}} = F_{\text{rec}}(\mathbf{W}_{\text{rec}}, \mathbf{Z}) \in \mathbb{R}^{B \times L \times C} \quad (18)$$

where $F_{\text{rec}}(\cdot)$ denotes the transformation process of the RecDecoder, with $\mathbf{W}_{\text{rec}}$ as its weight parameters. This process reconstructs the encoded features $\mathbf{Z}$ into the original input space, yielding $\tilde{\mathbf{X}} \in \mathbb{R}^{B \times L \times C}$. Similarly, the PreDecoder is represented as

$$\tilde{\mathbf{Y}} = F_{\text{pre}}(\mathbf{W}_{\text{pre}}, \mathbf{Z}) \in \mathbb{R}^{B \times l \times C} \quad (19)$$

where $F_{\text{pre}}(\cdot)$ denotes the transformation process of the PreDecoder, with $\mathbf{W}_{\text{pre}}$ as its weight parameters. The PreDecoder transforms the encoded features $\mathbf{Z}$ into a representation of predicted future values $\tilde{\mathbf{Y}} \in \mathbb{R}^{B \times l \times C}$.

Leveraging the F³ EM, we can obtain the embedding features $\mathbf{Z}$, reconstruction loss $\mathcal{L}_{\text{rec}}$, and prediction loss $\mathcal{L}_{\text{pre}}$ for the backend ASGM based on the original input data $\mathbf{X}'$, reconstructed data $\tilde{\mathbf{X}}$, and predicted data $\tilde{\mathbf{Y}}$

$$\mathcal{L}_{\text{rec}} = \left\{ e_r \in \mathbb{R}^N \mid e_r^i = \|\tilde{\mathbf{X}}_i - \mathbf{X}''_i\|_2 \right\} \quad (20)$$

$$\mathcal{L}_{\text{pre}} = \left\{ e_p \in \mathbb{R}^N \mid e_p^i = \|\tilde{\mathbf{Y}}_i - \mathbf{X}''_i\|_1 \right\} \quad (21)$$

where $\|\cdot\|_2$ and $\|\cdot\|_1$ represent the L2 norm and L1 norm, respectively. Notably, the triplet $\{\mathbf{Z}, e_r, e_p\}$ characterizes the input samples distinctly. By leveraging the F³ EM, anomalous instances exhibiting marked discrepancies in reconstruction and prediction can be discerned as deviating substantially from their normal counterparts.

C. Frequency-Domain Deviation Prior Module

The FDPM identifies anomalous patterns and irregularities in the frequency domain, enhancing insight extraction from spectral characteristics. Let $\mathbf{X}^L \in \mathbb{R}^{N \times L}$ denote the input data, we first apply the FFT to the input data to obtain the frequency representation

$$\mathbf{F} = \mathbb{H}(\mathbf{X}^L) \in \mathbb{C}^{N \times (L/2+1)}. \quad (22)$$

We then compute the amplitude spectrum $\mathcal{D}$

$$\mathcal{D} = |\mathbf{F}| \in \mathbb{R}^{N \times (L/2+1)} \quad (23)$$

where $|\cdot|$ represents the elementwise absolute value. To analyze the deviations in the frequency domain of $\mathbf{X}'$, we calculate the average amplitude spectrum, denoted by $\mathcal{D}_{\text{avg}} \in \mathbb{R}^{N \times (L/2+1)}$, by taking the median of the amplitude spectrum across all samples: $\mathcal{D}_{\text{avg}} = \text{Median}(\mathcal{D})$. The total deviation ξ_i of each data point from the average amplitude spectrum is then computed

$$\xi_i = \sum_{j=1}^{L/2+1} |\mathcal{D}_{i,j} - \mathcal{D}_{\text{avg},j}|. \quad (24)$$

Through this way, we obtain the frequency-domain deviation of all inputs $\Theta = [\xi_1, \xi_2, \dots, \xi_N]^T \in \mathbb{R}^N$. Lastly, the total deviations are standardized using a median-based normalization method, yielding $\tilde{\xi}_i$

$$\tilde{\xi}_i = \frac{\xi_i - \text{Median}(\Theta)}{\text{Std}(\Theta)} \quad (25)$$

where $\tilde{\xi}_i$ remains unaffected by extreme values or large scales, enhancing scalability invariance.

D. Anomaly Score Generation Module

The ASGM is a novel anomaly detection approach that learns a scoring function to distinguish between normal and anomalous data points using prior knowledge about anomalies. To achieve this, ASGM utilizes prior anomaly labels $\mathbf{X}_A$. ASGM first identifies pseudolabeled normal samples using the MZS algorithm, based on a composite error metric $e = e_r + e_p$. Let $\{\mathbf{Z}, e_r, e_p\}$ denote the output from the F³ EM. We obtain $\tilde{\mu} = \text{Median}(e)$ and $\tilde{\sigma} = \text{Median}(|e - \tilde{\mu}|)$ through e . The MZS calculation process can be expressed as

$$\mathbf{M} = \frac{\alpha * |e - \tilde{\mu}|}{\tilde{\sigma}} \quad (26)$$

where α is the 75th percentile of the standard normal distribution. We identify pseudolabeled normal data points $\mathbf{Z}^+$ that meet the condition $\mathbf{M} < 3\tilde{\sigma}$.

Next, we define $\mathbf{Z}^- \in \mathbb{R}^{n_2 \times z}$ as the prior anomaly label data $\mathbf{X}_A$ after embedding with F³ EM. Next, we form

$\{(\mathbf{Z} \in \mathbb{R}^{m \times z}, \mathbf{y} \in \mathbb{R}^m) | ((\mathbf{Z}^+, \mathbf{Z}^-), (y^+, y^-))\}$, where m is the total number of data points in $\mathbf{Z}$. Finally, we train a score model, represented by $\mathbf{S}$, using the dataset $\mathbf{Z}$. The score model is optimized via the gradient harmonizing mechanism (GHM) loss function, whose details will be provided in the training process. The score model can be represented as

$$\text{AnomalyScore} = \mathbf{S}(\mathbf{Z}, \mathbf{W}_s, \mathbf{b}_s) \quad (27)$$

where $\mathbf{W}_s$ and $\mathbf{b}_s$ are the learnable weights and biases, respectively.

E. Training Procedure

The FreAD framework is trained in two stages. In the first stage, FreAD employs FDPD to guide F^3 EM training, obtaining embedding features and modeling parameters. In the second stage, the pretrained F^3 EM is used to train the anomaly score model.

The F^3 EM is trained by minimizing the following loss function $\mathcal{L}$:

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \mathcal{L}_{\text{pre}} + \beta^{-1} * \mathcal{L}_{\text{prior}}. \quad (28)$$

The overall $\mathcal{L}$ consists of three components: $\mathcal{L}_{\text{rec}}$, $\mathcal{L}_{\text{pre}}$, and prior regularization term $\beta * \mathcal{L}_{\text{prior}}$. The prior regularization term is composed of two parts: the frequency-domain deviation coefficient β and the prior loss $\mathcal{L}_{\text{prior}}$.

The frequency-domain deviation coefficient β is calculated as

$$\beta = \max\{|\tilde{\xi}_i|, \epsilon\} \quad (29)$$

where $|\tilde{\xi}_i|$ is inversely proportional to β , which implies that inputs with smaller deviations are assigned larger weights, allowing the model to focus more attention on them. Meanwhile, regularization is performed through a small positive value ϵ to prevent overfitting and ensure a stable learning process.

The prior loss $\mathcal{L}_{\text{prior}}$ measures the dissimilarity between the latent representations $\mathbf{z}_i$ and the centroid $\mathbf{z}_M$ of the data obtained by applying PCA [21] to the geometric median [22] $\mathbf{X}_M$ of the original input data $\mathbf{X}'$

$$\mathcal{L}_{\text{prior}} = \frac{1}{N} \sum_{i=1}^N (1 - \text{cosine}(\mathbf{z}_i, \mathbf{z}_M)) \quad (30)$$

where $\text{cosine}(\cdot, \cdot)$ denotes the cosine similarity between two vectors. By minimizing $\mathcal{L}_{\text{prior}}$, the model is encouraged to learn latent representations that are close to the centroid, with larger weights $\tilde{\xi}_i^{-1}$ assigned to instances with higher similarity, the regularization term encourages the model to focus on important features, addressing the limitations of conventional Fourier transforms in capturing transient anomalies. Finally, the score model is trained using the pretrained F^3 EM to estimate the anomaly score of the input data, with the objective of minimizing the GHM loss function

$$\mathcal{L}_{\text{GHM}} = -\frac{1}{m} \sum_{i=1}^m \frac{y_i \log(\mathbf{S}(\mathbf{z}_i)) + (1 - y_i) \log(1 - \mathbf{S}(\mathbf{z}_i))}{\sum_{j=1}^m \mathbb{I}(g_j \in [b_i, b_{i+1}))} \quad (31)$$

where $g_j = |\delta(\mathbf{z}_j) - y_j|$ is the error for the j th sample, $\delta(\cdot)$ denotes the sigmoid activation function, b_i is the i th bin boundary,

and $\mathbb{I}(g_j \in [b_i, b_{i+1}))$ is an indicator function that equals 1 if g_j falls within the i th bin, and 0 otherwise.

IV. EXPERIMENTATION AND EVALUATION

A. Datasets Description

In this work, we evaluate FreAD's performance using three datasets for encrypted traffic anomaly detection in ICS: CTU13 [28], IDS2018 [25], and DoHBrw-2020 (DoH2020) [29]. The need for effective detection arises from the increasing reliance on communication networks, exposing critical infrastructure to sophisticated cyber threats. ICS environments include vital systems, such as power grids and manufacturing processes, where operational integrity is essential. Identifying anomalies in encrypted traffic is crucial for maintaining security and preventing catastrophic failures.

The CTU13 dataset, from Czech Technical University, features 13 simulated botnet traffic scenarios designed to reflect industrial network attacks, complete with detailed labeling and bidirectional NetFlow files. This dataset captures various botnet attacks in a university network, utilizing hypertext transfer protocol secure (HTTPS) and transport layer security (TLS) 1.3 protocols. The IDS2018 dataset, from the Canadian Institute of Cybersecurity, includes various simulated attack scenarios relevant to industrial systems, such as denial of service (DoS) and user to root attacks. This dataset encompasses a diverse range of attack types, including DDoS attacks and port scanning, and employs TLS and secure shell (SSH) encryption protocols, making it critical for ICS. We extract encrypted traffic to rigorously assess FreAD's effectiveness. Lastly, the DoH2020 dataset, also from the Canadian Institute of Cybersecurity, concentrates on domain name system (DNS) over HTTPS (DoH) traffic, featuring both benign and malicious patterns. This dataset simulates real-world encrypted DNS behaviors using DoH, which is valuable for constructing lightweight detection models. Given the rising adoption of DoH in industrial contexts for secure DNS resolution, this dataset offers a valuable testbed to evaluate FreAD's capability in detecting anomalies within modern encrypted protocols. By utilizing these datasets, we assess FreAD's anomaly detection capabilities, adopting a 2% anomaly ratio for realistic evaluation. A summary of the experimental data is presented in Table I. While these datasets provide a strong foundation for our evaluation, generalizing our results to other encryption protocols or industrial systems warrants further investigation.

B. Evaluation Metrics

We employed three widely recognized evaluation metrics to assess the performance of FreAD: AUC-ROC, AUC-PR, and F1-score. These metrics provide a comprehensive perspective on FreAD's anomaly detection capabilities in encrypted traffic flows. AUC-ROC measures the model's ability to distinguish between normal and anomalous data points; however, it can be misleading in situations of class imbalance, as a high score may not accurately reflect the model's true detection capability for the minority class (anomalies). In contrast, AUC-PR focuses on the model's precision and recall, making it more suitable for handling imbalanced data and offering a clearer insight

TABLE I
OVERVIEW OF DoH2020, IDS2018, AND CTU13 DATASETS WITH TRAINING AND TESTING SPLITS

DataSets	DoH2020				IDS2018				CTU13			
Dimensions	350				400				600			
	Train		Test		Train		Test		Train		Test	
Datasets Splitting	Normal	Anomalous	Normal	Anomalous	Normal	Anomalous	Normal	Anomalous	Normal	Anomalous	Normal	Anomalous
2%	294 000	6000	105 683	2157	345 842	7058	345 842	7058	48 580	992	48580	992
5%	285 000	15 000	105 683	5563	134 102	7058	134 102	7058	48 580	2557	48580	2557
10%	270 000	30 000	105 683	11 743	63 522	7058	63 522	7058	48 580	5398	48580	5398
15%	255 000	45 000	105 683	18 650	39 995	7058	39 995	7058	48 580	8573	48580	8573
20%	240 000	60 000	105 683	26 421	28 232	7058	28 232	7058	48 580	12 145	48 580	12 145

TABLE II
COMPARISON OF AUC-ROC, AUC-PR, AND F1-SCORE PERFORMANCE OF FREAD WITH VARIOUS COMPETING ALGORITHMS

Algorithms	DoH2020			IDS2018			CTU13			Average		
	AUCROC	AUCPR	F1	AUCROC	AUCPR	F1	AUCROC	AUCPR	F1	AUCROC	AUCPR	F1
ECOD [6]	0.960	0.010	0.486	0.997	0.699	0.643	0.732	0.034	0.471	0.896	0.248	0.533
COPOD [26]	0.991	0.053	0.488	0.917	0.104	0.628	0.642	0.025	0.469	0.850	0.061	0.528
INNE [28]	0.858	0.003	0.473	0.988	0.310	0.641	0.945	0.171	0.592	0.930	0.161	0.569
IForest [27]	0.998	0.171	0.489	0.999	0.898	0.642	0.794	0.047	0.521	0.930	0.372	0.551
OCSVM [20]	0.845	0.003	0.477	<u>0.997</u>	0.748	0.641	0.739	0.038	0.511	0.860	0.263	0.543
Deep SVD [29]	0.756	0.502	0.502	<u>0.989</u>	<u>0.963</u>	<u>0.977</u>	0.807	0.455	0.227	0.851	0.640	0.569
Dev-Net [10]	0.982	0.955	0.766	0.976	0.944	0.955	0.979	<u>0.945</u>	0.867	<u>0.979</u>	<u>0.948</u>	0.863
OmniAnomaly [13]	0.994	0.855	0.922	0.999	0.663	0.798	0.834	<u>0.362</u>	0.533	<u>0.942</u>	0.627	0.751
Anomaly-Transformer [18]	0.994	0.855	<u>0.974</u>	0.931	0.679	0.823	0.967	0.901	<u>0.959</u>	0.965	0.812	<u>0.919</u>
DCdetector [19]	0.973	0.734	0.852	0.900	0.158	0.316	0.934	0.861	0.916	0.936	0.584	0.695
USAD [15]	<u>0.998</u>	0.804	0.891	0.678	0.105	0.312	<u>0.990</u>	0.904	0.949	0.889	0.604	0.717
DAEMON [12]	0.999	<u>0.987</u>	0.994	0.999	0.312	0.475	0.999	0.999	0.999	0.999	0.766	0.823
FreAD (Proposed)	0.999	0.994(0.7%↑)	<u>0.974</u>	0.999	0.999(3.6%↑)	0.999(2.2%↑)	0.999	0.999	0.999	0.999(2%↑)	0.997(4.9%↑)	0.991(7.2%↑)

Bold indicates the highest performance for each metric. Green underlined denotes the second highest performance. These color codings help to easily identify the top-performing algorithms at a glance.

into anomaly detection performance. The F1-score balances precision and recall, ensuring that even in environments with scarce anomalies, the model can effectively capture these rare events. Through these metrics, we demonstrate FreAD's efficacy in detecting anomalies.

C. Baselines and Parameter Settings

Baselines: We compared FreAD's performance with 12 competing algorithms: ECOD [6], COPOD [26], OCSVM [20], INNE [28], IForest [27], OmniAnomaly [13], USAD [15], DAEMON [12], Dev-Net [10], Deep SVD [29], Anomaly-Transformer [18], and DCdetector [19]. We used default parameters as described in the literature. Traditional algorithms were implemented using Pyod, while FreAD was built with PyTorch. Experiments were run on a computer with a GeForce RTX 3090 GPU and an Intel Xeon Silver 4216 CPU, using PyCharm 2022.2.5.

Parameter settings: The FDTL and FDCL components in FreAD's F³ EM utilize Kaiming initialization and zero biases, with fused features consisting of 1024 dimensions and embedding features comprising 128 dimensions. The extended decoder's reconstruction and prediction branches feature 1024-dimensional hidden layers, employing a 0.05 dropout rate and LayerNorm normalization, initialized with Xavier normal initialization. The output dimensions for the reconstructed input and predicted output match the input dimensions and consist of 50 dimensions, respectively. The number of X_A is set to 30. The training process incorporates early stopping with a patience of five epochs, using the AdamW optimizer with a weight decay of $1e-5$. The learning rate is adjusted through CosineAnnealingLR, starting at $1e-4$, with a minimum learning

rate of $1e-7$ over 200 epochs. The batch size is set to 128, and λ is fixed at 0.01. In addition, the regularization coefficient ϵ is set to 0.001. For hyperparameter selection, we employed a manual tuning approach to optimize critical parameters, including the learning rate, dropout rate, and batch size, which allowed us to systematically identify the most effective settings for FreAD's performance. In addition, the ASGM module is structured as a three-layer MLP with dimensions 64, 32, and 1, also utilizing a 0.05 dropout rate and LayerNorm normalization.

D. Experiment Results With Competitive Methods

The experimental results compiled in Table II unequivocally demonstrate that our proposed FreAD algorithm significantly outperforms 12 competing methods across multiple benchmark datasets (DoH2020, IDS2018, and CTU13). By evaluating performance through AUC-ROC, AUC-PR, and F1-score metrics, we comprehensively assessed the algorithm's effectiveness in anomaly detection. Across all evaluation metrics, FreAD consistently surpassed competing methods. The average results reveal remarkable achievements: an AUC-ROC of 0.999 (representing 2%↑ over DAEMON), an AUC-PR of 0.997 (4.9%↑), and an F1-score of 0.991 (7.2%↑) over the nearest competitor. These substantial improvements underscore FreAD's reliability and adaptability across diverse data types and anomaly patterns.

Dataset-specific analysis provides deeper insights into the algorithm's performance. On the DoH2020 dataset, FreAD and DAEMON shared the highest AUC-ROC score of 0.999, while FreAD distinctly led with the highest AUC-PR of 0.994 (0.7%↑) and the best F1-score of 0.974. This demonstrates the algorithm's precision in anomaly detection while maintaining an exceptional balance between precision and recall.

TABLE III
EVALUATION OF THE IMPORTANCE OF EACH COMPONENT IN F^3 EM

Components	DoH2020			IDS2018			CTU13		
	AUCROC	AUCPR	F1	AUCROC	AUCPR	F1	AUCROC	AUCPR	F1
FreAD _{FDTL}	0.999	0.886	0.876	0.999	0.993	0.997	0.999	0.995	<u>0.996</u>
FreAD _{FDCL}	0.999	0.923	0.957	0.998	0.996	0.997	0.999	0.983	<u>0.994</u>
FreAD _{Rec}	0.996	0.774	0.822	0.999	0.983	0.962	0.999	<u>0.997</u>	0.994
FreAD _{Pre}	<u>0.997</u>	0.868	0.842	0.999	0.964	0.945	0.999	0.999	0.982
FreAD _{FDPM}	0.999	<u>0.986</u>	<u>0.967</u>	0.999	<u>0.998</u>	<u>0.998</u>	0.999	0.995	0.992
FreAD (Proposed)	0.999	0.994	0.974	0.999	0.999	0.999	0.999	0.999	0.999

Bold indicates the highest performance for each metric. Green underlined denotes the second highest performance. These color codings help to easily identify the top-performing algorithms at a glance.

The IDS2018 dataset further validated FreAD's capabilities. It achieved the top AUC-ROC of 0.999, the highest AUC-PR of 0.999 (3.6%↑), and the best F1-score of 0.999 (2.2%↑ over the second method). These results highlight FreAD's exceptional performance in complex, highly variable environments, where maintaining precise detection remains challenging. On the CTU13 dataset, FreAD again distinguished itself by securing the highest scores across all metrics: an AUC-ROC of 0.999, an AUC-PR of 0.999, and an F1-score of 0.999. This performance reflects the algorithm's robust capability in handling intricate anomaly detection scenarios with complex attack patterns. When comparing FreAD to other methods, DAEMON emerged as the closest competitor, particularly in AUC-ROC and AUC-PR metrics. However, FreAD consistently outperformed DAEMON, especially in AUC-PR and F1-score, emphasizing its superior precision-recall balance. Alternative approaches, such as OmniAnomaly and Anomaly-Transformer, demonstrated intermittent success on specific datasets but failed to match FreAD's consistent performance. Traditional methods, such as IForest and OCSVM, significantly lagged behind, further highlighting FreAD's innovative approach.

E. Ablative Analysis

1) Evaluating the Necessity of Each Component in F^3 EM:

In this comprehensive evaluation, we investigate the critical contributions of each component within the F^3 EM in the FreAD framework. By systematically removing key component: FDTL, TDCL, reconstruction (Rec), prediction (Pre), and FDPM. Table III illustrates their respective impact on algorithm performance, with subscripts indicating the removed components.

The DoH2020 dataset reveals profound performance variations when components are absent. Eliminating the FDTL component precipitates a dramatic decline: AUC-PR plummets from 0.994 to 0.886, while the F1-score correspondingly drops from 0.974 to 0.876. Similarly, removing the FDCL component significantly degrades performance, with AUC-PR falling to 0.923 and F1-score reducing to 0.957. These results unequivocally underscore the indispensable role of temporal and channel characteristics in anomaly detection. The IDS2018 dataset further corroborates the significance of these components. FreAD_{Rec} demonstrates notably inferior performance when the reconstruction decoding branch is omitted, registering AUCPR and F1-score of 0.983 and 0.962, respectively. Likewise, FreAD_{Pre} exhibits suboptimal performance without its prediction branches, with an AUCPR of 0.964 and F1-score of 0.945. On the CTU13 dataset, the critical nature of reconstruction and prediction branches becomes even more evident. Removing the

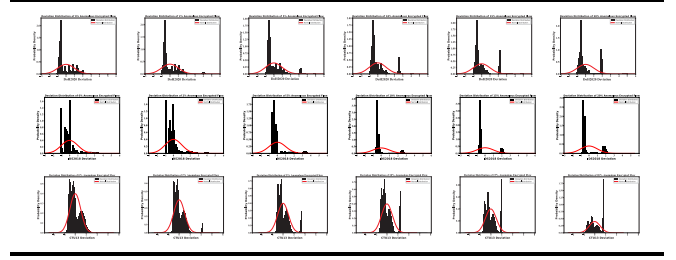


Fig. 3. Probability density distributions of deviation priors from FDPM across varying contamination rates. (The black curve represents the deviation distribution and the red curve represents the normal distribution.)

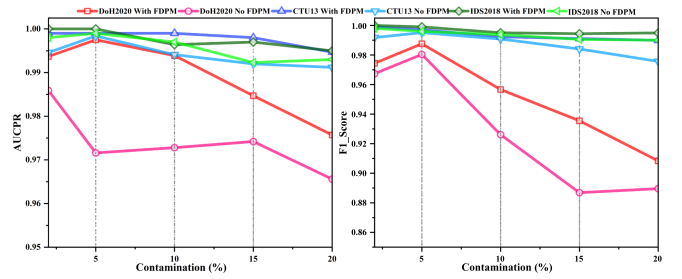


Fig. 4. Comparative analysis of AUC-PR and F1-scores for the FreAD with and without FDPM on three datasets.

reconstruction module diminishes the F1-score to 0.982, while eliminating the prediction module causes AUC-PR to decline to 0.994 and F1-score to 0.994. This powerfully illustrates the profound value of reconstruction and prediction capabilities in deciphering data structures, identifying anomalous patterns, and anticipating potential risks.

Notably, the absence of the FDPM module substantially compromises algorithmic performance. Particularly on the DoH2020 dataset, the performance degradation is stark: AUC-PR decreases to 0.986 and the F1-score drops to 0.967. This highlights the distinctive advantages of frequency-domain bias priors in augmenting anomaly detection capabilities.

2) Evaluating the Effectiveness of FDPM Under Varying Contamination Rates: Fig. 3 illustrates the probability density distribution of encrypted flow deviations generated by the FDPM module across three datasets, including a control group with 0% contamination for comparison. As the contamination rate increases, anomalous deviations begin to emerge, increasingly resembling the normal deviation distribution. Notably, at lower contamination rates, these deviations remain distinct, providing valuable prior information for FreAD. Fig. 4 complements this analysis by comparing the performance of the FreAD framework with and without the FDPM across the same datasets. The results clearly demonstrate that incorporating FDPM significantly enhances anomaly detection performance, particularly as the contamination rate rises. This is evidenced by improvements in AUC-PR and F1-scores, highlighting the effectiveness of FDPM in distinguishing between normal and anomalous samples. The performance on the DoH2020 dataset showcases the substantial benefits of incorporating FDPM into the FreAD framework.

TABLE IV
PERFORMANCE OF FREAD WITH DIFFERENT CLASS-IMBALANCED LOSS FUNCTIONS

Different Loss	DoH2020			IDS2018			CTU13			Average Metrics		
	AUCROC	AUCPR	F1	AUCROC	AUCPR	F1	AUCROC	AUCPR	F1	AUCROC	AUCPR	F1

DoH2020: 0.999, 0.999, 0.974, 0.999, 0.999, 0.999, 0.999, 0.999, 0.999, 0.999, 0.999, 0.999, 0.999 (1.1%) 0.991 (2.2%)
 Focal: 0.999, 0.969, 0.922, 0.999, 0.958, 0.971, 0.999, 0.998, 0.997, 0.999, 0.988, 0.963
 Dice: 0.999, 0.981, 0.941, 0.999, 0.987, 0.973, 0.999, 0.999, 0.993, 0.999, 0.982, 0.952
 WeightCE: 0.999, 0.921, 0.913, 0.999, 0.968, 0.962, 0.999, 0.993, 0.992, 0.999, 0.961, 0.956

Bold indicates the highest performance for each metric. Green underlined denotes the second highest performance. These color codings help to easily identify the top-performing algorithms at a glance.

Notably, at a contamination rate of 20%, the FDP-enabled framework achieves exceptional AUC-PR and F1-score of 0.976 and 0.908, respectively, significantly surpassing the non-FDPM approach (0.965 and 0.889). This marked improvement underscores the robustness of our approach in detecting anomalies, even in the presence of high contamination rates. Moreover, the FDP-enabled framework exhibits a more gradual decline in performance as the contamination rate increases, demonstrating its ability to adapt to changing data conditions and maintain its effectiveness in anomaly detection.

Similar trends are observed in the performance on the CTU13 and IDS2018 datasets, where the FDP-enabled framework consistently outperforms its counterparts, achieving higher AUC-PR and F1-score across all contamination rates. On CTU13, the FDP-enabled framework achieves exceptional scores, with AUC-PR and F1-score of 0.998 and 0.991 at 15% contamination and 0.994 and 0.989 at 20% contamination, surpassing the non-FDPM methods. Similarly, on IDS2018, the FDP-enabled framework attains AUC-PR and F1-score of 0.997 and 0.994 at 15% contamination and 0.995 and 0.994 at 20% contamination, outperforming the non-FDPM approach. The consistent improvements across these three datasets underscore the benefits of integrating FDP into the FreAD framework, enabling it to effectively detect anomalies in diverse data environments and adapt to changing data conditions.

3) Evaluating FreAD With Different Class-Imbalanced Loss Functions: Table IV compares the performance of the FreAD framework using various class-imbalanced loss functions across three datasets, with a focus on GHM loss, which has proven highly effective for anomaly detection.

The FreAD framework utilizing GHM loss achieves impressive metrics, with AUCROC, AUCPR, and F1-score values nearing 1.0 across all datasets. Specifically, it records an AUCROC of 0.999, an AUCPR of 0.999, and an F1-Score of 0.974 on the DoH2020 dataset. The average metrics for GHM loss show improvements of 1.1% in AUCPR and 2.2% in F1-score compared to other loss functions.

In contrast, other loss functions perform lower. For example, the FreAD framework using focal loss achieves an AUCPR of 0.969 and an F1-score of 0.922 on DoH2020, while the framework with dice loss reaches an AUCPR of 0.981 and an F1-score of 0.941. These results highlight a consistent performance gap compared to GHM loss. In addition, the framework with WeightedCE loss exhibits inferior performance across all metrics.

The superior performance of the FreAD framework with GHM loss is attributed to its effective handling of class imbalance and outliers. By dynamically adjusting class weights,

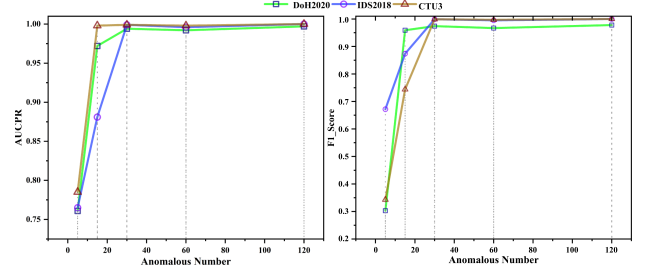


Fig. 5. Comparative analysis of AUCPR and F1-score performance for the FreAD with varying numbers of abnormal prior labels on three datasets.

TABLE V
MODEL PERFORMANCE AND COMPLEXITY ANALYSIS

Methods	Datasets(Dim)	Memory Usage(MB)	Max Memory Usage(MB)	Running Time(ms)	Inference Speed(FPS)	Flops	Parameters
Deep SVD	DoH2020 (350)	8.13	8.20	41.005	390.20	11.82 Kmac	11.78 k
	IDS2018 (400)	8.13	8.21	41.000	390.24	14.42 Kmac	13.38 k
	CTU13 (600)	8.13	8.25	43.000	372.09	19.92 Kmac	19.31 k
Dev-Net	DoH2020 (350)	45.67	516.34	123.390	129.67	0.56 Gmac	1.23 M
	IDS2018 (400)	48.12	518.56	146.359	109.82	0.64 Gmac	1.23 M
	CTU13 (600)	52.89	530.12	187.771	85.21	0.96 Gmac	1.23 M
OmniAnomaly	DoH2020 (350)	32.45	512.78	113.491	140.98	1.24 Gmac	2.56 M
	IDS2018 (400)	36.12	520.45	130.623	122.49	1.42 Gmac	2.56 M
	CTU13 (600)	48.78	540.23	145.085	110.28	2.13 Gmac	2.56 M
Anomaly Transformer	DoH2020 (350)	548.15	932.30	201.642	79.35	1.66 Gmac	4.75 M
	IDS2018 (400)	711.27	1194.32	216.017	74.07	1.9 Gmac	4.75 M
	CTU13 (600)	1592.16	2598.22	274.000	58.39	2.85 Gmac	4.75 M
DCDetector	DoH2020 (350)	97.85	146.84	160.000	100.00	186.88 Mmac	884.74 k
	IDS2018 (400)	127.66	180.83	163.000	98.16	212.73 Mmac	896.26 k
	CTU13 (600)	272.12	345.16	168.024	95.22	316.13 Mmac	942.34 k
USAD	DoH2020 (350)	8.15	9.51	62.000	258.06	541.71 Kmac	322.61 k
	IDS2018 (400)	8.15	9.91	63.000	253.97	708.1 Kmac	422.1 k
	CTU13 (600)	8.16	12.00	67.000	238.81	1.59 Mmac	948.15 k
DAEMON	DoH2020 (350)	3.26	26.91	164.000	97.56	1.75 Gmac	3.72 M
	IDS2018 (400)	3.65	27.82	167.001	95.81	1.84 Gmac	3.74 M
	CTU13 (600)	5.21	34.46	179.000	89.39	2.18 Gmac	3.82 M
FreAD	DoH2020 (350)	8.49	555.69	64.830	246.80	125.94 Mmac	126.06 M
	IDS2018 (400)	8.13	728.98	78.004	205.12	164.46 Mmac	164.61 M
	CTU13 (600)	8.13	1632.67	83.000	192.77	369.86 Mmac	370.22 M

GHM loss allows the framework to focus on the most informative samples, enhancing overall performance. These findings confirm that GHM loss is particularly well suited for the FreAD framework, significantly improving its anomaly detection effectiveness compared to the other evaluated loss functions.

4) Evaluating the Impact of the Amount of Anomalous Prior Data: Fig. 5 assesses the impact of the number of anomaly priors on the results. The experimental findings indicate that FreAD performs effectively across three different datasets, achieving impressive outcomes with only a small number of anomaly priors. Notably, the algorithm can reach near-perfect performance with just 30 anomaly labels. As the number of anomaly prior data points increases, both AUCPR and F1-score quickly approach 1.0, reflecting excellent precision, recall, and overall detection ability. This quick convergence is consistent across all three datasets, underscoring the algorithm's strong generalization potential. In addition, it demonstrates high data efficiency, consistency across datasets, and rapid learning capabilities, making FreAD especially valuable for real-world anomaly detection. It can effectively learn from a limited number of anomaly examples, positioning it as a powerful and adaptable solution for network security and intrusion detection challenges.

5) Model Performance and Complexity Analysis: As shown in Table V, we compares FreAD with seven DL-based anomaly detection frameworks across the DoH2020 (350), IDS2018 (400), and CTU13 (600) datasets. To enhance real-time performance, we employ batch processing with a batch size of 16. We analyzed six key metrics: memory usage (MB), max memory

usage (MB), running time (ms), inference speed (FPS), floating point operations per second (FLOPs), and parameters.

FreAD's average memory usage is 8.49, 8.13, and 8.13 MB across the datasets, comparable to deep SVD and USAD, but significantly lower than Dev-Net, OmniAnomaly, and Anomaly-Transformer. However, its peak memory usage, reaching 555.69, 728.98, and 1632.67 MB on the CTU13 dataset, indicates challenges in temporary storage during high-dimensional data processing. Despite this, FreAD's low average memory usage makes it suitable for resource-constrained ICS environments. In terms of performance, FreAD achieves competitive running times of 64.83, 78.00, and 83.00 ms, with inference speeds of 246.80, 205.12, and 192.77 FPS. While slightly slower than deep SVD and USAD, it significantly outperforms Dev-Net, OmniAnomaly, Anomaly-Transformer, and DAEMON, demonstrating its capability to meet real-time requirements for encrypted traffic anomaly detection. With computational complexities of 125.94, 164.46, and 369.86 Mmac, and parameter counts of 126.06, 164.61, and 370.22 M, FreAD shows robust feature extraction capabilities but requires higher hardware resources. Its efficient inference speed suggests potential optimization through hardware acceleration (e.g., GPU or TPU).

Despite challenges with high peak memory usage and elevated FLOPs, FreAD's advantages in average memory usage, running time, and inference speed make it suitable for real-time encrypted traffic anomaly detection in ICS. On high-performance devices, its computational efficiency enhances detection capabilities, while hardware acceleration can further optimize resource demands, improving FreAD's adaptability across various operational environments.

V. CONCLUSION

In this article, we present FreAD, a novel frequency-domain semisupervised anomaly detection framework designed to tackle the challenges of detecting anomalies in encrypted network traffic within ICS. By utilizing frequency-domain features and prior anomaly knowledge, FreAD effectively alleviates the representation information bottleneck associated with the point-by-point mapping approach used in traditional DL-based anomaly detection models. Extensive experiments show that FreAD achieves state-of-the-art AUC-ROC, with average improvements of 4.9% in AUC-PR and 7.2% in F1-score. FreAD offers a promising solution to the critical issue of anomaly detection in ICS, essential for maintaining the reliability and security of these systems. Future work will explore transfer learning and few-shot learning to adapt FreAD to new anomaly types. We will also assess its performance in real ICS environments, expanding beyond the current three benchmark datasets to ensure generalization across various encryption protocols and systems.

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